

Week 2 Control a character - paper route answers

Teacher only. Keep this sheet away from pupil copies.

GRID 1, reset script. Under when green flag clicked: go to x: 0 y: 0, then point in direction 90. Rows 3, 4 and 5 stay empty. Order matters less here than in the controls, but the pupil should not add a change block. This grid is copied, not created. Never count it as authorship of a control.

GRID 2, right control. Under when right arrow key pressed: change x by 10. One block only. Rows 2, 3 and 4 stay empty. The word right should be ringed in the hat. A pupil on the Independent Task who used change x by ___ and wrote a positive number of their own has met the check; record the number they chose.

GRID 3, left control. Under when left arrow key pressed: change x by -10. One block only. The minus sign must be present and visible. If the pupil writes 10 and then corrects it to -10 after Trace 1, that is a successful debug - record it as such, it is stronger evidence than getting it right first time.

GRID 4, up control, Challenge route. Under when up arrow key pressed: change y by 10. The common error is change x by 10. That is a wrong-axis error, not a wrong-sign error - name it precisely in the record.

GRID 5, down control, Challenge route. Under when down arrow key pressed: change y by -10.

WORD BANK. The bank deliberately offers all four motion blocks in every grid, plus the two reset blocks. The choice is the evidence. A pupil who picks change y by 10 for the right arrow has made a real, readable error - record what they chose and what prompt corrected it. A bank containing only the right answer would prove nothing.

RINGING THE ARROW. Ringing, underlining, pointing to, or saying the arrow name all count. This is the paper equivalent of setting the dropdown on the Events block. If a pupil cannot mark the page, accept them pointing to the arrow word while you ring it, and record physical access help.

MAZE, the two marked squares. GREEN FLAG SQUARE is $x = 0$, $y = 0$. That is where the green flag puts Courier this week, and where every trace starts. MAZE HOME is $x = -180$, $y = -120$, the player's start square in the real maze. Pupils sometimes confuse the two - a counter that starts in the wrong place gives a whole wrong column, so check the start square before the trace, not after.

MAZE, corridor count. Ten right presses. The counter passes -170, -160, -150, -140, -130, -120, -110, -100, -90 and stops on -80. $x = -80$, $y = -120$ is the last free square. An eleventh press would land on $x = -70$, which is the near edge of the wall that runs from $x = -70$ to $x = -50$. The correct answer to the last row is "wall - stop".

MAZE, return count. Ten left presses take the counter back to $x = -180$, $y = -120$. Same number as the way out. Ask why, and accept: minus ten undoes plus ten. That is the core idea of the week said in the pupil's own words.

MAZE, why the finish flag is out of reach this week. The finish is $x = 180$, $y = 120$. The wall from $x = -70$ to $x = -50$ runs all the way up from the floor to $y = 80$. Right and left alone cannot get past it. The counter must first go up to $y = 90$ to clear it. From $y = -120$ that is 21 up presses. Counting the rows on the printed grid is a correct method - the pupil does not have to physically make 21 moves. This is the point that motivates all four controls in Week 4.

MAZE, beacon. $x = 20$, $y = 20$ is open ground in the middle corridor. So are $x = 10$ and $x = 20$ on the $y = 0$ row. Nothing in this week's tracing touches a wall except the deliberate corridor stop.

TRACE 1 answers. Row 3: $x = 10$, $y = 0$. Row 4: $x = 20$, $y = 0$. Last row: the wrong block is change x by 10; it should say change x by -10. Expect and accept the pupil's own wording, for example "it needs a minus" or "it should be take away ten".

TRACE 1, what the diagnosis must show. The pupil must separate the two halves. The hat is right - the left key really did fire. The block underneath is wrong - it moves the counter the wrong way. A pupil who says only "it is broken" has not diagnosed it; ask which block and wait. A pupil who points at the change block and then at the minus sign on the repair card has diagnosed it.

PREDICT AND CHECK answers. Left: expect -10, counter lands on -10. Then right: expect 0, counter lands on 0. y stays 0 throughout. Both ticks. A pupil who predicted wrongly and then wrote the true traced value has done the task correctly - the column is called I expect, not I was right. Never ask a pupil to change a prediction after tracing.

TRACE 2 answers. $x = 10$, then 20, then 10, then 0. $y = 0$ on every row. The finish is the green flag square, back where it started. This is the week's success check.

TRACE 3 answers. Right press 2: $x = -160$, $y = -120$, free. Right press 3: $x = -150$, $y = -120$, free. Right press 10: $x = -80$, $y = -120$, free. Right press 11 would land on: $x = -70$, $y = -120$, wall - stop. Left presses back to MAZE HOME: 10.

CHALLENGE A answers. right, right, up, up finishes at $x = 20$, $y = 20$ - tick. up, right, up, right finishes at $x = 20$, $y = 20$ - tick. Any own order with two rights and two ups also finishes at $x = 20$, $y = 20$. The explanation to accept: there is nothing in the way here, so the presses add up to the same total whichever order they come in. In the real maze the order would matter, because a wall would stop one of the orders. That contrast is worth drawing on the printed maze if the pupil is ready.

CHALLENGE B answers. Four right presses of 5 reach $x = 20$, so the prediction should be 4. The rows read 5, 10, 15, 20 with $y = 0$ throughout. After the second green flag: 5, 10, 5, 0. The finish is 0 again, the same as with tens - the counter just takes smaller steps. Check the pupil left grids 2 and 3 at 10 and -10; this investigation is done in the table only.

WHAT COUNTS WHEN A PUPIL POINTS. Pointing to the correct block in the word bank while an adult writes it into the grid is a correct answer, and the adult writes "pupil pointed, adult scribed" in the support record. Pointing to the wrong block in the faulty script, then to the minus sign on the repair card, is a complete

diagnosis. Pointing to the square the counter should land on, before it is moved, is a prediction.

WHAT COUNTS WHEN A PUPIL DRAWS. An arrow drawn from one square to the next on the maze is a recorded move. A drawn arrow pointing right beside change x by 10, and left beside change x by -10, is a correct explanation of the minus sign. Numbers written straight onto the maze squares instead of in the table are acceptable - transcribe them into the table yourself and say so.

WHAT COUNTS WHEN A PUPIL SAYS IT. Record exact words, not a tidied version. "Minus goes backwards" is a correct explanation. "It goes the other way" is correct. "It is wrong" is not yet an explanation - offer a general prompt such as "which way does it go?" and wait before modelling.

COMMON ERRORS AND WHAT THEY MEAN. Wrong axis (a y block under a left or right arrow) means the pupil has not yet separated x from y - reteach with the two arrows drawn on the maze edges, do not just correct it. Wrong sign (change x by 10 under the left arrow) is this week's taught bug - if the pupil then finds and repairs it themselves, that is the check met, not a mistake. Sliding the counter two squares for one press means press-and-release has not landed - restate as "one press is one square" and retrace from the green flag square. Forgetting to reset before the second test gives a finish of 40 instead of 20 in Challenge A - this is the same error a held key causes on screen, and is worth naming.

THE READY-MADE CONTROLS CARD. This is the paper equivalent of W02_Recovery_Horizontal.sb3. Hand it over only with an adult, and only when a pupil cannot otherwise take part. Record "inherited code" in the support record and keep the missing script as a named next step, exactly as you would with the recovery file. A pupil who copied the card has not created a control.

COMPLETING THE RECORD. Mark the five checks from what you saw the pupil do, not from which route sheet they held. The pupil signs off their own statements by ticking, pointing, or telling you which ones are true. Record the route as "paper" in this teacher guide. Do not write on, change or reprint the AQA summary sheet: Week 2 claims no outcome and there is nothing to enter. Carry any missing second script forward as a named next step; all four pupil-created controls are still due in the main maze in Week 4.

What this week evidences

Week 2 assessment destination, in the pack's own words: Foundation practice. "These lessons build foundations for AQA 71638 Programming with Scratch (unit 6), Level One. Neither Week 1 nor Week 2 completes the full unit." No AQA outcome is claimed this week. The work here evidences the pack's own core completion statement for Week 2: "The pupil creates and tests BOTH right and left scripts. One working script is an early milestone and leaves a recorded next step. All four controls are checked in the main maze in Week 4." The shared goal is: "I can use key events to move a sprite and fix a direction mistake." This feeds outcome 2, "Create keyboard arrow-control scripts", which is claimed in Week 4 and not before.

Known limit of the paper route

One of the pack's five checks differs in kind between the two routes, and the teacher record must say so. Check 5 is "Save, locate and reopen the controls." On paper the pupil names the pack W02_MyControls, dates it, files it, finds it again, rereads both grids and retraces one control to the same finish. That evidences what the check is for: the pupil keeps their own work, finds it again without help, and confirms their scripts are still there and still behave the same. It does not evidence the Scratch operation itself - File, Save to your computer; an .sb3 file on disk; File, Load from your computer. A pupil on the paper route has not done those things, and a teacher must not record that they have. Two reasons this does not leave the week incomplete. First, Week 2 claims no AQA outcome; the pack states plainly that neither Week 1 nor Week 2 completes the unit, so no assessed evidence is lost. Second, none of the six 71638 outcomes is a save-and-reopen outcome. The outcome Week 2 feeds is outcome 2, "Create keyboard arrow-control scripts", and creating and testing both control scripts is exactly what the paper route evidences. So the honest record for check 5 on this route reads "named, filed, found and retraced the paper controls", and that is a complete record of what was observed. A record that instead read "saved and reopened a .sb3" would be the false mapping this route exists to avoid, because nobody reading it later could tell it was untrue. One much smaller difference, named for completeness and costing nothing: a pupil on paper never meets key auto-repeat, so the misconception that a held key moves the sprite several times never arises for them. The press-and-release rule is still taught, as "one press is one square".